Chapter 2: Motion in One Dimension

These notes describe motion along a straight line. We track where an object is, how far it has moved, and how quickly its position changes. Every quantity comes with a unit, and checking units is the fastest way to catch a mistake in a calculation.

2.1 Position and Displacement

Position is measured from a chosen origin. When a cart moves from an initial position x_0 to a final position x_f, its displacement is the change in position:

$$\Delta x = x_f - x_0$$

Displacement has a direction. A cart that ends up behind its starting point has a negative displacement, even though the distance it travelled is positive.

Units

Positions and displacements are measured in metres. A displacement of $3\,\text{m}$ to the left is written as $-3\,\text{m}$ when the positive direction points right.

2.2 Speed and Velocity

Average velocity is displacement divided by the time taken. For a trip that lasts Δt, the average velocity is $v = \Delta x / \Delta t$, and its unit is metres per second.

$$v = \frac{\Delta x}{\Delta t}$$

If we know the kinetic energy of the cart and its mass, its speed follows from

$$v = \sqrt{\frac{2K}{m}}$$

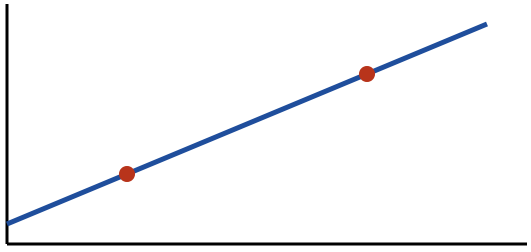


Figure 2.1 Position against time for a cart moving at constant velocity.

2.3 Acceleration

Acceleration is the rate at which velocity changes. A cart that speeds up from rest to $6\,\text{m/s}$ in two seconds has an average acceleration of $3\,\text{m/s}^2$.

$$a = \frac{\Delta v}{\Delta t}$$

When the acceleration is constant, the velocity grows by the same amount every second and a graph of velocity against time is a straight line.

Free fall

Near the surface of the Earth, a dropped ball has a nearly constant downward acceleration g as long as air resistance is small. Timing a fall over a known height is a simple way to estimate it in the lab.